

3.3 Martingales

Definition (Martingale). A sequence ξ_n is a **martingale** with respect to a filtration $(\mathcal{F}_n)$ if

1. $\xi_n \in L^1$.
2. (ξ_n) is adapted to $(\mathcal{F}_n)$.
3. $E(\xi_{n+1}|\mathcal{F}_n) = \xi_n$ a.s.

Example 3.3. Let (η_n) be independent random variables with $E(\eta_n) = 0$ for every $n \geq 1$. Define

$$\xi_n = \sum_{k=1}^n \eta_k, \quad \mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n).$$

Then (ξ_n) is a martingale with respect to $(\mathcal{F}_n)$.

Proof. The sum ξ_n is $\mathcal{F}_n$ -measurable and integrable, since

$$E(|\xi_n|) \leq \sum_{k=1}^n E(|\eta_k|) < \infty.$$

Since η_{n+1} is independent of $\mathcal{F}_n$, we have

$$\begin{aligned} E(\xi_{n+1}|\mathcal{F}_n) &= E(\xi_n + \eta_{n+1}|\mathcal{F}_n) \\ &= E(\xi_n|\mathcal{F}_n) + E(\eta_{n+1}|\mathcal{F}_n) \\ &= \xi_n + E(\eta_{n+1}) = \xi_n. \end{aligned}$$

□

Example 3.4. Let $\xi \in L^1$, and let $(\mathcal{F}_n)$ be a filtration. Then the sequence defined by

$$\xi_n = E(\xi|\mathcal{F}_n)$$

is a martingale with respect to $(\mathcal{F}_n)$.

Proof. ξ_n is integrable, $\mathcal{F}_n$ measurable by definition of conditional expectation. Furthermore, since $\mathcal{F}_n \subset \mathcal{F}_{n+1}$, we have

$$E(\xi_{n+1}|\mathcal{F}_n) = E(E(\xi|\mathcal{F}_{n+1})|\mathcal{F}_n) = E(\xi|\mathcal{F}_n) = \xi_n.$$

□

Exercise 3.3. Show that a martingale (ξ_n) has constant expectation:

$$E(\xi_n) = E(\xi_1) \quad \text{for all } n \geq 1.$$

Proof. By definition of martingale and conditional expectation, we have

$$E(\xi_{n+1}) = E(E(\xi_{n+1}|\mathcal{F}_n)) = E(\xi_n).$$

By induction, we have $E(\xi_n) = E(\xi_1)$ for all $n \geq 1$. □

Exercise 3.4. Suppose (ξ_n) is a martingale with respect to $(\mathcal{F}_n)$. Show that it is also martingale with respect to its natural filtration $(\mathcal{G}_n)$ defined by

$$\mathcal{G}_n = \sigma(\xi_1, \dots, \xi_n).$$

Proof. Since $\mathcal{G}_n \subset \mathcal{F}_n$, we have

$$\begin{aligned} E(\xi_{n+1}|\mathcal{G}_n) &= E(E(\xi_{n+1}|\mathcal{F}_n)|\mathcal{G}_n) \\ &= E(\xi_n|\mathcal{G}_n) = \xi_n. \end{aligned}$$

□

Exercise 3.5 (Symmetric Random Walk). Let (η_n) be independent random variables with

$$P\{\eta_n = 1\} = P\{\eta_n = -1\} = \frac{1}{2},$$

and define the symmetric random walk by

$$\xi_0 = 0, \quad \xi_n = \sum_{k=1}^n \eta_k, \quad n \geq 1.$$

Show that $\xi_n^2 - n$ is a martingale for the filtration

$$\mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n), \quad \mathcal{F}_0 = \{\emptyset, \Omega\}.$$

Proof. Since $|\xi_n| \leq n$, the variable $\xi_n^2 - n$ is integrable, and it is $\mathcal{F}_n$ -measurable. The new step is independent of $\mathcal{F}_n$ and satisfies

$$E(\eta_{n+1}) = 0 \quad \text{and} \quad E(\eta_{n+1}^2) = 1.$$

Thus,

$$\begin{aligned} E(\xi_{n+1}^2 - (n+1)|\mathcal{F}_n) &= E((\xi_n + \eta_{n+1})^2 - (n+1)|\mathcal{F}_n) \\ &= E(\xi_n^2 + 2\xi_n\eta_{n+1} + \eta_{n+1}^2|\mathcal{F}_n) - (n+1) \\ &= \xi_n^2 + 2\xi_n E(\eta_{n+1}) + E(\eta_{n+1}^2) - (n+1) \\ &= \xi_n^2 - n. \end{aligned}$$

□

Exercise 3.6. $\zeta_n = (-1)^n \cos(\pi\xi_n)$ is a martingale.

Proof. Homework.

□

Definition. A sequence (ξ_n) is a **submartingale** with respect to a filtration $(\mathcal{F}_n)$ if

1. $\xi_n \in L^1$.
2. (ξ_n) is adapted to $(\mathcal{F}_n)$.
3. $E(\xi_{n+1}|\mathcal{F}_n) \geq \xi_n$ a.s.

Similarly, (ξ_n) is a **supermartingale** with respect to $(\mathcal{F}_n)$ if

1. $\xi_n \in L^1$.
2. (ξ_n) is adapted to $(\mathcal{F}_n)$.
3. $E(\xi_{n+1}|\mathcal{F}_n) \leq \xi_n$ a.s.

Exercise 3.7. Suppose (ξ_n) is a martingale¹ with $\xi_1 \in L^2$. Show that (ξ_n^2) is a submartingale with respect to the same filtration.

¹From now on, we consider that martingale (ξ_n) is adapted to its natural filtration $(\mathcal{F}_n)$.

Proof. Since $\xi_n \in L^2$, its square is integrable. It is clear that

$$\xi_n^2 \text{ is } \mathcal{F}_n\text{-measurable for every } n \geq 1.$$

Conditional Jensen, applied to $x \mapsto x^2$, gives

$$E(\xi_{n+1}^2 | \mathcal{F}_n) \geq (E(\xi_{n+1} | \mathcal{F}_n))^2 = \xi_n^2.$$

□

Indeed, the result of Exercise 3.7 can be generalized to any convex function φ . That, is, if $\varphi(\xi_n) \in L^1$, then $(\varphi(\xi_n))$ is a submartingale.

3.4 Games of Chance

Definition. A sequence (α_n) is **previsible** with respect to a filtration $(\mathcal{F}_n)$ if α_n is $\mathcal{F}_{n-1}$ -measurable for every $n \geq 1$.

In gambling, α_n is the stake for round n , and (α_n) is called a **gambling strategy**. Let us consider following example:

1. (η_n) : winnings per unit stake in game n .
2. $\xi_n = \eta_1 + \dots + \eta_n$: total winnings after n games with $\xi_0 = 0$.
3. $\mathcal{F}_n := \sigma(\eta_1, \dots, \eta_n)$, with $\mathcal{F}_0 = \{\emptyset, \Omega\}$.

The game is said to be ...

1. **fair** if $E(\xi_n | \mathcal{F}_{n-1}) = \xi_{n-1}$ a.s.
2. **favorable** if $E(\xi_n | \mathcal{F}_{n-1}) \geq \xi_{n-1}$ a.s.
3. **unfavorable** if $E(\xi_n | \mathcal{F}_{n-1}) \leq \xi_{n-1}$ a.s.

Suppose you may vary your stake from game to game. Under a strategy (α_n) , your total winnings are

$$\zeta_0 = 0, \quad \zeta_n = \sum_{k=1}^n \alpha_k (\xi_k - \xi_{k-1}).$$

Proposition 3.1. *Let (α_n) be previsible strategy such that each α_n is bounded and let (ζ_n) be the winnings defined above. If (ξ_n) is a martingale, then (ζ_n) is a martingale for the same filtration. If also $\alpha \geq 0$ for every n , then a supermartingale (ξ_n) yields supermartingale (ζ_n) , and a submartingale (ξ_n) yields a submartingale (ζ_n) .*

Proof. Each term in ζ_n is $\mathcal{F}_n$ -measurable. Boundedness of the stakes also gives

$$E(|\alpha_k(\xi_k - \xi_{k-1})|) \leq \|\alpha_k\|_\infty E(|\xi_k - \xi_{k-1}|) < \infty.$$

So every ζ_n is integrable.

Since α_n and ζ_{n-1} are $\mathcal{F}_{n-1}$ -measurable, the identity

$$\zeta_n = \zeta_{n-1} + \alpha_n(\xi_n - \xi_{n-1})$$

implies

$$\begin{aligned} E(\zeta_n | \mathcal{F}_{n-1}) &= \zeta_{n-1} + \alpha_n(E(\xi_n | \mathcal{F}_{n-1}) - \xi_{n-1}) \\ &= \zeta_{n-1}. \end{aligned}$$

□